

# GROW NOW, BREATHE LATER?

## Economic Development and the Exposure to Air Pollution

PIP V

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### Executive Summary

Fine particulate matter (PM<sub>2.5</sub>) exposure represents a major threat to public health, contributing to around a million premature deaths per year. In this paper, we investigate whether economic development can simultaneously act as clean air policy by testing the Environmental Kuznets Curve hypothesis against PM<sub>2.5</sub> specifically, a pollution standard introduced after Grossmann and Krueger's foundational 1995 paper. Using panel data from 174 countries, ranging from 2000 to 2019 drawn from the World Bank's World Development Indicators, we run a log-log quadratic model of population-weighted PM<sub>2.5</sub> exposure on GDP per capita. A naïve OLS model finds only that richer countries are cleaner through a linear decline. Once spatial and year fixed effects absorb unobserved heterogeneity, an inverted U-shape relationship emerges. We find that, within countries, PM<sub>2.5</sub> exposure increases until a turning point of \$7,201 (with a wide confidence interval) after which it decreases. From the chosen mediators, the regressions only support for the significance of renewable energy share and trade percentage. Given that the initial increase of PM<sub>2.5</sub> affects vulnerable groups disproportionately, government should take precautions to the weakest.

# I. Introduction

Poor air quality is a reality for most humans on Earth. An estimated 7.3 billion people are subject to annual average PM2.5<sup>1</sup> concentrations exceeding the World Health Organization's safe limits (Rentschler & Leonova, 2023). Such exposure to pollution is more than a mere inconvenience; it poses a real threat to public health. Roughly 1 million premature deaths per year were attributable to short-term PM2.5 exposure between 2000 and 2019, representing a staggering 2.08% of global deaths annually (Yu et al., 2024). Thus, there is an obvious need for global change. What measures may alleviate this state of affairs? Beyond regulation, there is at least one other candidate that may address this crisis: economic development. That is, if economic development were associated with decreases in PM2.5 exposure, development policy would simultaneously act as clean air policy. Therefore, this paper examines whether, and to what extent, GDP per capita predicts annual population-weighted exposure to fine particulate matter (PM2.5) across countries. We test the Environmental Kuznets Curve logic specifically against PM2.5, building on the foundational work by Grossmann and Krueger (1995) while deviating in our choice of environmental indicators. At the time of publication, Grossmann and Krueger did not have access to the now dominant PM2.5 standard, as it was only introduced in 1997, two years after their publication (U.S. EPA, 1997). Summing up, this paper revisits the Environmental Kuznets Curve more than 30 years later with modern data and slightly different environmental indicators.

## II. Theory and Mechanisms

Simon Kuznets' (1955) seminal work found that the interaction between economic development and income inequality can be captured by an inverted U-shape. That is, with economic development, income inequality first increases until a turning point, after which it decreases. Such inverted U-shape relationships are referred to as *Kuznets curves*, honoring the author. Towards the end of the 20<sup>th</sup> century, Grossman & Krueger (1995) examined the relationship between economic development and environmental degradation, observing that in poor countries, air and water quality worsen with economic development but then improve after "some critical level of income has been reached" (Grossman & Krueger, 1995, p. 370). Given the similar nature of the relationship and U-shaped curve, Grossman and Krueger's findings have subsequently been dubbed the *Environmental Kuznets curve*. Such relationships reflect that the environmental benefits of economic development may be only reaped in the long run, as early-stage growth initially leads to environmental degradation before higher income levels from growth trigger mechanisms that reverse it.

What follows from this insight is that development must be understood beyond the perspective of simple income growth. As economies develop, sectoral shifts structurally affect the allocation of labor and capital and affect both the spatial construct of the economy and environmental outcomes linked to economic activity through a series of growth-related mechanisms. This paper investigates whether such a non-linear relationship originally identified by Kuznets and later applied to environmental outcomes by Grossman and Krueger, holds for population-weighted PM2.5 exposure across 174 countries between 2000 and 2019, and which mechanisms account for it.

While the exact mechanism of the interplay between economic development and exposure to pollution does not follow a linear pathway and depends on a long series of external factors, the literature reflects that pollution exposure tends to change structurally at different stages of development. During the early stages of development, pollution-generating mechanisms dominate structural transformation. Since early stages of economic growth depend heavily on manufacturing and energy-intensive production, PM2.5 concentration drastically increases due to manufacturing processes and fossil fuel combustion. As Antweiler et al. (2001) show, economic growth mechanically increases pollution through the scale

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<sup>1</sup> A dominant standard for fine atmospheric particles.

effect. The underlying mechanism of this effect is that a proportional expansion of economic activity generates a proportional increase in pollution. This is an interesting baseline mechanism because it works even if the composition of output and pollution intensity is held constant, meaning that an upward pressure on pollution exists independently of a structural labor change from agriculture to manufacturing. Particularly in the early stages of development, this effect is strong as output volume changes before structural sectoral changes occur.

Next, looking closer at structural transformation in early development, the second mechanism operates through the composition effect. Structural transformation initially moves the labor share and capital out of low-polluting agricultural activity into high-polluting manufacturing and industrial production. Following higher productivity and output of manufacturing jobs in the early stages, workers naturally shift to the higher-income sector. Consequently, the increased labor share leads to a growth in output in high-polluting industries (Copeland & Taylor, 2004; Grossman & Krueger, 1995). Using a constant elasticity of substitution Utility Framework, Jaeger et al. (2023) provide an updated EKC of the inverted U-shape relationship between air pollutants and income across countries, consistent with composition driven dynamics in early stages of development as explained above.

At higher income levels, the pollution-increasing mechanisms not only start to flatten off, but a set of counter-directional mechanisms gain dominance. This ultimately bends the Environmental Kuznets curve downwards once an income-level-driven turning point is reached.

If an economy reaches the tipping point of structural transformation where a sectoral move from manufacturing to services occurs, the composition effect reverses. First, in the early stages of growth the composition effect mechanically leads to an increase in exposure to PM<sub>2.5</sub>. A compositional shift of the labor share from manufacturing to the service sector creates exactly the opposite effect. Post-industrial transformation thereby shifts labor and capital away from pollution-intensive industries towards less polluting service sectors (Antweiler et al., 2001; Copeland & Taylor, 2004). The aggregate output of pollution, such as PM<sub>2.5</sub>, consequently coincides with the share of GDP produced by manufacturing industries. Second, as firms and the government become richer and gain access to more advanced and environmentally friendly technologies, pollution per unit of output decreases. Renewable energy, particle filters, or more energy-efficient machinery, for example, enable a decrease in pollution, without lowering levels of output. This mechanism is part of what Antweiler et al. (2001) introduce as the “technique effect”. Third, rising income also creates a demand-side pressure on environmental regulation policies as higher income levels increase willingness to pay (WTP) for cleaner air and thus the demand for environmental regulation. Ultimately, this increase in political pressure and available resources from the adjusted WTP provides governments both with the incentive and fiscal capacity to enforce stricter pollution regulations (Beckerman, 1992; Copeland & Taylor, 2004). Lastly, increased trade openness from economic growth triggers a series of mechanisms that reduce exposure to pollutants within a country. One prominent mechanism that Copeland and Taylor (2004) discuss is the pollution haven hypothesis, which reflects the idea that when higher-income countries liberalize trade, they also outsource pollution-intensive production processes to lower-income countries with weaker environmental standards. Consequently, PM<sub>2.5</sub> exposure decreases with economic growth even if the production processes themselves do not adapt.

Following a sectoral shift, the urban population density increases as workers migrate to manufacturing centers. Looking at the example of China’s urbanization wave between 2000 and 2010, Aunan & Wang (2014) show that increased urbanization raises population-weighted PM<sub>2.5</sub> exposure as more people live closer to emission sources such as traffic, construction, or industrial zones. Interestingly, this exposure mechanism increases PM<sub>2.5</sub> exposure even if per-capita emission does not increase proportionally, reflecting the important nuance in this paper that exposure to PM<sub>2.5</sub> does not follow proportionally from PM<sub>2.5</sub> production or other emissions such as CO<sub>2</sub>. What the explained mechanisms show is that economic development not only leads to an increased exposure to PM<sub>2.5</sub> through an increase in pollution output, but also through spatial urban effects.

### III. Data

Our dataset consists of seven different series from the World Bank World Development Indicators (WDI) database, covering the period 2000-2019 (20 years, annual). The unit of observation is the country-year. The panel includes 174 countries with a total of 3,185 observations. The dependent variable is a mean annual population-weighted exposure to fine particulate matter, which is measured in micrograms per cubic meter ( $\mu\text{g}/\text{m}^3$ ). The World Bank draws on satellite-derived estimates produced through the Global Burden of Disease study, which ensures broad spatial coverage as it also includes countries with limited ground-level monitoring infrastructure. Our primary explanatory variable is GDP per capita, which is expressed in current US dollars. We have included four mediating variables to isolate the mechanism effects through which economic development influences air pollution exposure. All variables are drawn from the WDI and are measured annually. (i) Urban population share is expressed as a percentage of total population and captures the spatial concentration of economic activity. We expect that a higher urbanization rate increases the independent variable. (ii) Industry value added as a share of GDP is a mediator on the income-pollution channel, capturing the scale effect of structural change. Manufacturing and energy-intensive are primary sources of PM2.5 emissions. (iii) Trade Openness is measured as total trade as a percentage of GDP and works as a proxy for a country's integration into global value chains. The pollution haven hypothesis is what motivates its inclusion: greater trade openness may allow higher-income countries to offshore pollution-intensive production to lower-income trade partners. This may potentially bias the GDP coefficient downward if omitted. (iv) Renewable energy share is expressed as a percentage of total energy consumption and reflects the cleanliness of a country's energy mix. Since the combustion of fossil fuels is a major source of fine particulate matter, a higher share of renewable energy is expected to reduce PM2.5 exposure.

The WDI database download initially included 199 country-level entities; however, aggregate regional observations, such as "Sub-Saharan Africa" and "High income", were removed as they would cause double-counting. A further reduction was done by dropping all observations with missing GDP or PM2.5 data in any year across the full 2000-2019 window. 174 countries are therefore observed. Due to some control variables missing, especially renewable energy share and trade openness data among lower-income countries, the resulting panel of 3,185 observations falls short of the theoretical maximum of 3,480 (174 countries x 20 years). The estimation sample is held constant at 3185 observations across all six models.

| Variable                    | Unit                     | N     | Mean      | SD        | Min    | Median   | Max        |
|-----------------------------|--------------------------|-------|-----------|-----------|--------|----------|------------|
| PM2.5 exposure              | $\mu\text{g}/\text{m}^3$ | 3,185 | 26.80     | 16.73     | 5.13   | 22.12    | 107.14     |
| GDP per capita              | current USD              | 3,185 | 12,891.09 | 19,009.01 | 136.93 | 4,264.66 | 123,678.70 |
| Urban population share      | %                        | 3,185 | 58.57     | 22.14     | 11.64  | 60.85    | 100.00     |
| Industrial production share | % of GDP                 | 3,185 | 27.65     | 12.40     | 3.05   | 25.23    | 86.67      |
| Trade openness              | % of GDP                 | 3,185 | 85.64     | 47.99     | 15.28  | 76.08    | 437.33     |
| Renewable energy            | % of energy              | 3,185 | 31.63     | 28.84     | 0.00   | 22.80    | 98.30      |

*Table 1: Descriptive statistics*

Descriptive statistics for all key variables are reported in their original, non-log units in Table 1. The mean PM2.5 exposure across observed countries is  $26.8 \mu\text{g}/\text{m}^3$ , substantially above the WHO guideline of  $5 \mu\text{g}/\text{m}^3$ . It is considerably dispersed with a standard deviation of 16.7 and a maximum of  $107.1 \mu\text{g}/\text{m}^3$  that reflects the severe pollution levels observed in parts of South Asia and Sub-Saharan Africa. The mean GDP per capita of \$12,891 masks an extremely wide distribution as the standard

deviation of \$19,009 exceeds the mean itself. This underscores the importance of log-transformation. Trade openness shows the widest relative variation of any control, ranging from 15.3% to 437.3% of GDP. This large range reflects the inclusion of highly open, small economies alongside large, closed ones.

## IV. Regression analysis

### IV.I Model Setup

The baseline specification is a log-log quadratic model following Kuznets (1955) and Koster (2025). This is specifically chosen for a multitude of reasons. First and most importantly, it allows us to interpret coefficients as elasticities. Second, the observed PM2.5 levels are heavily right-skewed, which would dominate the fit. Additionally GDP per capita levels vary between 136\$ and 123,678\$, a range in which relative changes are more useful.

$$\ln(PM2.5_{it}) = \alpha_i + \tau_t + \beta_1 \cdot \bar{x}_{it} + \beta_2 \cdot \bar{x}_{it}^2 + \delta Z_{it} + \varepsilon_{it}, \quad \text{with } \bar{x}_{it} = \ln(GDP_{pc})_{it} - \bar{x}_{mean}$$

Here  $i$  denotes the 174 countries and  $t$  the years 2000-2019. Where  $\bar{x}_{it} = \ln(GDP/cap)_{it} - \bar{x}$  is the mean-centered log income,  $\alpha_i$  are country fixed effects, and  $\tau_t$  year fixed effects. In order to get an inverted-U relationship, we need  $\beta_1 > 0$  and  $\beta_2 < 0$  and the turning point  $-\beta_1/(2\beta_2)$  to lie within the observed income range. The regression moves through six different models: Model (1) is a pooled OLS regression that serves as the cross-sectional baseline, pooling all within- and between-country variation. As it does not account for country-specific or time-specific factors, it is likely subject to omitted variable bias and is included primarily as a benchmark. Model (2) introduces country fixed effects ( $\alpha^i$ ), absorbing all time-invariant country characteristics such as geography, climate, and long-run institutional quality. Identification now relies solely on within-country variation over time, which constitutes a substantially more credible test of the income–PM2.5 relationship. Model (3) adds year fixed effects ( $\gamma_t$ ) instead, capturing global shocks common to all countries in a given year, such as recessions or worldwide energy price shifts. Model (4) replaces these year dummies with a continuous linear time trend, a more parsimonious specification. Model (5) combines both country and year fixed effects, constituting the preferred total effect specification: it eliminates the two primary sources of omitted variable bias simultaneously and identifies the EKC relationship from within-country, within-year income variation alone. Finally, Model (6) extends Model (5) with a vector of time-varying control variables  $Z^i_t$ . The control variables are urban share, industry share of GDP, trade openness, and renewable energy production share. We additionally need to motivate the fact that we choose a mean-centered metric before squaring the term with  $\bar{x}_{it} = \ln(GDP_{pc})_{it} - \bar{x}_{mean}$ . Centering is necessary because the raw log and its square correlate at approximately 0.995, which would severely inflate standard errors and make the constant coefficient uninterpretable. Centering leaves the curvature parameter ( $\beta_2$ ) and the turning point unchanged.

## IV.II Results

The results of the regression models are presented in Table 2 and generated with the fixest library in R. The standard error is given as a clustered result.

| Dependent Variable:       | ln_pm25              |                      |                      |                      |                      |                      |
|---------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Model:                    | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
| <i>Variables</i>          |                      |                      |                      |                      |                      |                      |
| Constant                  | 3.140***<br>(0.043)  |                      |                      | -8.017<br>(4.936)    |                      |                      |
| ln GDP/cap (centred)      | -0.227***<br>(0.024) | -0.114***<br>(0.019) | -0.234***<br>(0.026) | -0.232***<br>(0.025) | 0.020<br>(0.032)     | 0.003<br>(0.032)     |
| (ln GDP/cap) <sup>2</sup> | -0.011<br>(0.013)    | -0.029***<br>(0.005) | -0.009<br>(0.014)    | -0.009<br>(0.013)    | -0.022***<br>(0.005) | -0.015***<br>(0.005) |
| year                      |                      |                      |                      | 0.006**<br>(0.002)   |                      |                      |
| Urban %                   |                      |                      |                      |                      |                      | 0.001<br>(0.002)     |
| Industry %GDP             |                      |                      |                      |                      |                      | 0.000<br>(0.001)     |
| Trade %GDP                |                      |                      |                      |                      |                      | -0.001*<br>(0.000)   |
| Renewables %              |                      |                      |                      |                      |                      | -0.003**<br>(0.002)  |
| <i>Fixed-effects</i>      |                      |                      |                      |                      |                      |                      |
| cc                        |                      | Yes                  |                      |                      | Yes                  | Yes                  |
| year                      |                      |                      | Yes                  |                      | Yes                  | Yes                  |
| <i>Fit statistics</i>     |                      |                      |                      |                      |                      |                      |
| Observations              | 3,185                | 3,185                | 3,185                | 3,185                | 3,185                | 3,185                |
| R <sup>2</sup>            | 0.334                | 0.965                | 0.341                | 0.337                | 0.969                | 0.970                |

Clustered (cc) standard-errors in parentheses

Signif. Codes: \*\*\*, 0.01, \*\*, 0.05, \*, 0.1

Table 2: PM2.5 exposure and economic development

The OLS Model (1) estimates a highly significant linear coefficient of -0.227, which also represents the income elasticity of the PM2.5 parameter. This implies that if GDP per capita increases by one percent, the mean exposure to pollutants decreases by 0.227 percent. The quadratic coefficient, on the other hand, is not significant. Richer countries are simply cleaner if this model is taken literally. This linear relationship can be observed within Figure 1. The constant or intercept of our regression can be interpreted as the exposure of ln(PM2.5) at the mean income ( $\ln(3.140) \approx 23 \mu\text{g}/\text{m}^3$ ).

Model (2) accounts for spatial fixed effects, meaning we are looking at the development over time within a specific country. This changes the result quite a bit. While the linear coefficient stays highly significant, the income elasticity term is now -0.114 and thus almost half of the basic OLS. The quadratic term, on the other hand, is now also highly significant with a coefficient of -0.029. Additionally of interest is the fact that the R<sup>2</sup> term changed from 0.334 to 0.965. Most of the unobserved variance missing in model (1) can thus be explained through unobserved spatial factors.

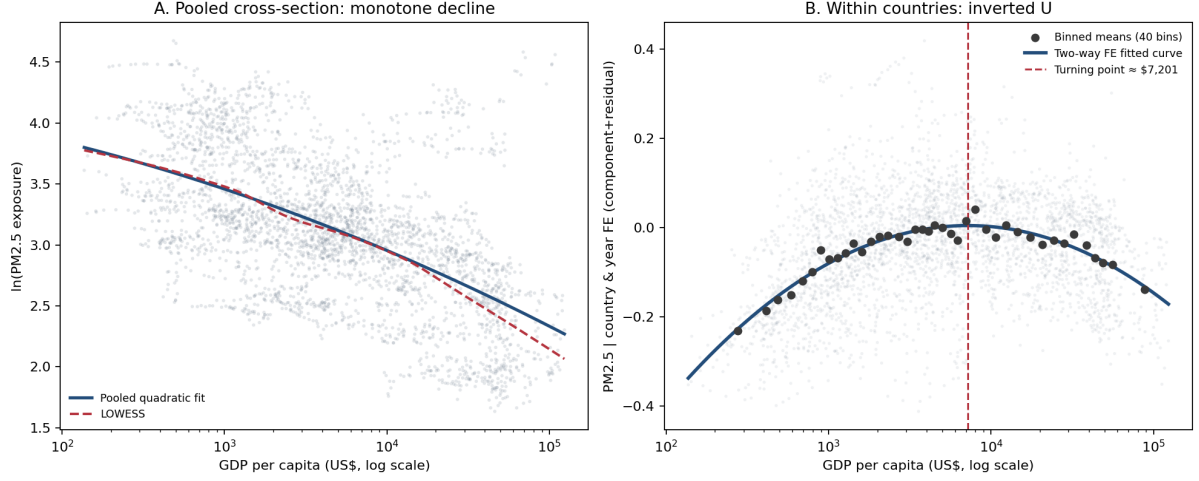


Figure 1:  $\ln(\text{PM2.5 exposure})$  and  $\ln(\text{GDP per capita})$  for Model (1) and Model (2)

Models (3) and (4) look at the temporal effects in a different way with very similar results. Model (3) looks at year fixed effects and Model (4) introduces a continuous time trend variable. Neither approach greatly increases the explanatory power of the model. Within both approaches, the constant remains highly significant while the quadratic term stays insignificant. The continuous time variable identifies a significant yearly increase of PM2.5 exposure of 0.6 %.

Model (5) accounts for both spatial and temporal fixed effects and is thus the best model to look at when investigating the total effect. Here, the squared term is the only significant value with  $\beta_2 = -0.022$ . The turning point can be calculated to be at \$7,201 per capita. However, it is important to note that the overall 95 percent confidence interval is set between \$1,626 and \$31,898, implying that many factors are at play in determining the exact point in economic development at which exposure to pollutants drops. The big confidence interval can be explained by the flat curvature surrounding the turning point and the standard error of the coefficients. Model (6) lastly considers several mediating variables. The variables urbanisation, industry share, trade openness, and share of energy production through renewable sources absorb parts of the curvature, with  $\beta_2$  dropping to -0.014. Even when accounting for mediating variables, the quadratic coefficient remains significant: the EKC survives. Both the trade share and the percentage of renewable energy production coefficients are negatively correlated with the dependent variable while being significant at  $p < 0.1$  and  $p < 0.05$ , respectively. This is interpretable as the mechanism through which economic development affects air pollution with trade and renewable energy being the most relevant.

### IV.III Functional Form Test

To support our quadratic log-log specification of the regression, we need to consider the possibility that a differently shaped curve is better suited to represent the relationship between our variables. To do that, we add a cubic variation to the two-way fixed effects model. The regression result yields a  $\beta_3 \approx 0.0015$  with  $p \approx 0.53$ , which means our quadratic specification holds up against an additional polynomial. To additionally ensure that no functional form misspecification in the form of emitted variables or missing interactions occurs, we employ a RESET-style test (Ramsey, 1969). After adding the squared fitted values of model (5), the result of  $p \approx 0.22$  does not confirm a potential misspecification.

### IV.IV Relation to Theory and Mechanisms

The regression results are consistent with the Environmental Kuznets Curve and broadly support the theoretical framework developed in Section II. The regression implies that PM2.5 exposure initially rises with economic development, before declining after reaching the turning point. The finding suggests

that the relationship between PM2.5 exposure and economic development is non-linear and cannot be described by a simpler growth-environment trade-off.

When urbanization, industry share, trade openness, and renewable energy use are included in Model (6), the weight of the EKC curvature decreases. This result is consistent with the mechanisms of structural transformation discussed in section II: part of the income-pollution relationship operates through modifications in economic structure rather than in income alone. Moreover, the reduction in the quadratic GDP coefficient provides indirect evidence for the composition effect outlined in Section II. However, because neither urbanization nor industry share are statistically significant, the results cannot provide direct evidence that these variables affect exposure to PM2.5. The coefficient for renewable energy share is negative and statistically significant, providing direct support for the technique effect discussed in Section II, where higher-income economies reduce their pollution through cleaner technologies. Consistent with Antweiler et al. (2001), countries with cleaner energy mixes experience lower PM2.5 exposure. The negative coefficient on trade openness is consistent with the trade-related mechanisms discussed in Section II and may reflect pollution haven dynamics. The theory suggests that advanced economies may reduce domestic pollution exposure by relocating pollution-intensive production abroad. The overall findings provide strong support for the technique effect, whilst offering more indirect evidence for the composition effect and trade-related channels outlined in Section II.

## V. Relation to policy

Let us present some policy considerations at the country-level and in the international sphere. We found that within a country, increases in GDP per capita are associated with an initial increase in PM2.5 exposure until a turning point, followed by a decrease in the long run (Fig. 1). So far, so good. Changes in pollution exposure, however, are only one of various processes associated with economic development. On one hand, PM2.5 exposure will increase in the short run, straining public health and, by extension, a country's health care system. Thus, in the short run, life expectancy can decrease due to increased pollution exposure. On the other hand, Preston (1975) stressed that economic development remains a strong predictor for life expectancy, a claim that has been supported by recent research (Prados De La Escosura, 2023). Economic development, he found, is associated with increases in life expectancy, captured by a concave curve with diminishing returns. Weighing these two developments against each other, Wang et al. (2022) find that population-weighted PM2.5 exposure can negatively affect increases in life expectancy stemming from increases in GDP per capita. Complicating the picture even further, Jorgenson et al. (2021) find that the negative relationship between life expectancy and PM2.5 exposure is more pronounced in countries with high levels of income inequality. Thus, if one agrees with Kuznets' (1955) claim that income inequality first increases with economic development, citizens face a double burden of both increasing pollution and increasing income inequality<sup>2</sup>, leading to a "substantial" (Jorgenson et al., 2021, p. 1) reduction in life expectancy, a burden disproportionately affecting those worst off. In a similar vein, Jorgenson et al. argue that 1) power is concentrated with the wealthy who exert influence on pollution regulation, 2) those worst off tend to live in areas with higher PM2.5 concentration, and 3) those disadvantaged typically experience worse health consequences from PM2.5 exposure because of other stressors they are subject to. In other words, pollution is a question of social (in-)justice and needs appropriate addressing to protect disadvantaged groups throughout economic transformation processes. Following John Rawls' intuition that social and economic inequalities should be arranged "to the greatest benefit of the least advantaged" (Rawls, 1971, p. 302), simply accepting life expectancy reductions for those worst off is a clear violation of his maximin approach to social justice. That is, a development policy that worsens the standing of vulnerable citizens in the short run can be deemed unjust, even if their position improves in the long run compared to

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<sup>2</sup> This is not to say that the turning points of income equality and pollution are the same, but rather that there may be some overlap.

baseline well-being. Thus, a Rawlsian state would intervene in an early stage of economic development to tackle the burden of the initial increase. Given the heterogeneity of developing countries, such interventions may take various forms of mitigation or adaptation. First, two obvious pathways towards mitigation include regulation and decoupling economic development from pollution through technological development. The success of regulation will heavily depend on enforceability and political feasibility. For example, restricting pollution-heavy production processes requires enforcement through inspections and credible penalties. Additionally, there may exist trade-offs between pollution regulation and economic development. That is, if pollution regulation hampers development in some significant way, this provides an argument against regulation and political feasibility is expected to be low. Decoupling approaches are dependent on access to technology that is not always available in developing countries. State- or NGO-driven development cooperation can, for example, provide access to less pollution-intensive machinery that would allow mitigating the initial pollution increase expected from industrialization. Such a leap-frogging strategy, however, requires local workers to have relevant skill sets and foreign technology to be compatible with local systems. In terms of adaptation, the impacts of air pollution on health can be limited by providing access to healthcare to both treat pollution-related illness and target structural vulnerabilities. For example, ensuring access to clean water and vaccines does not target pollution itself, but can positively impact life expectancy and ideally compensate for the decrease in life expectancy from pollution exposure. There is one important factor setting countries apart in how they may respond to the challenges of pollution: In densely populated countries, citizens are subject to higher average PM<sub>2.5</sub> exposure. This structural condition is very costly to change because of path dependencies and, therefore, does not represent a feasible mitigation strategy. In this context, adaptation strategies (as discussed above) are more advisable.

Let us now turn to the international sphere. In the long run and across countries, increases in GDP per capita are associated with decreases in PM<sub>2.5</sub> exposure. That is, with economic development, our results suggest a global reduction in pollution exposure. Therefore, it may seem that economic growth policy would simultaneously lead to improved global air quality (in the long run). At least one issue complicates the picture. As Grossman and Krueger (1995) concede, there is the possibility of pollution havens, that is, developed countries outsource pollution to even poorer countries. While the authors do not identify this development as a major factor in decreasing a given country's pollution, it remains morally salient because it fits into a bigger pattern of exploitation, often in the form of imperialist and/or neo-colonial trade patterns. Frame (2022) offers a model for understanding such "ecological imperialism" (inspired by World Systems Theory): the author characterizes international trade as structurally unequal. Developed *core* countries engage in ecologically unequal exchange with developing (*semi-*)*periphery* countries. In this relationship, the Global North not only extracts resources from the Global South but also offshores pollution and other types of environmental degradation. Circling back to the Rawlsian intuition of protecting those worst off, we can say that, on an international level, those worst off (the *periphery*) will disproportionately suffer from the *core*'s transformation processes, including the outsourcing of pollution. Tackling this unequal exchange is particularly troublesome in the international sphere since, in the absence of a global government, reform in this area depends on countries' willingness to cooperate. Given the ubiquity of pollution offshoring (Choi et al., 2025; Cole et al., 2021), a structural global departure from this practice will at the very least be contentious if not politically unfeasible.

## VI. Sensitivity analysis

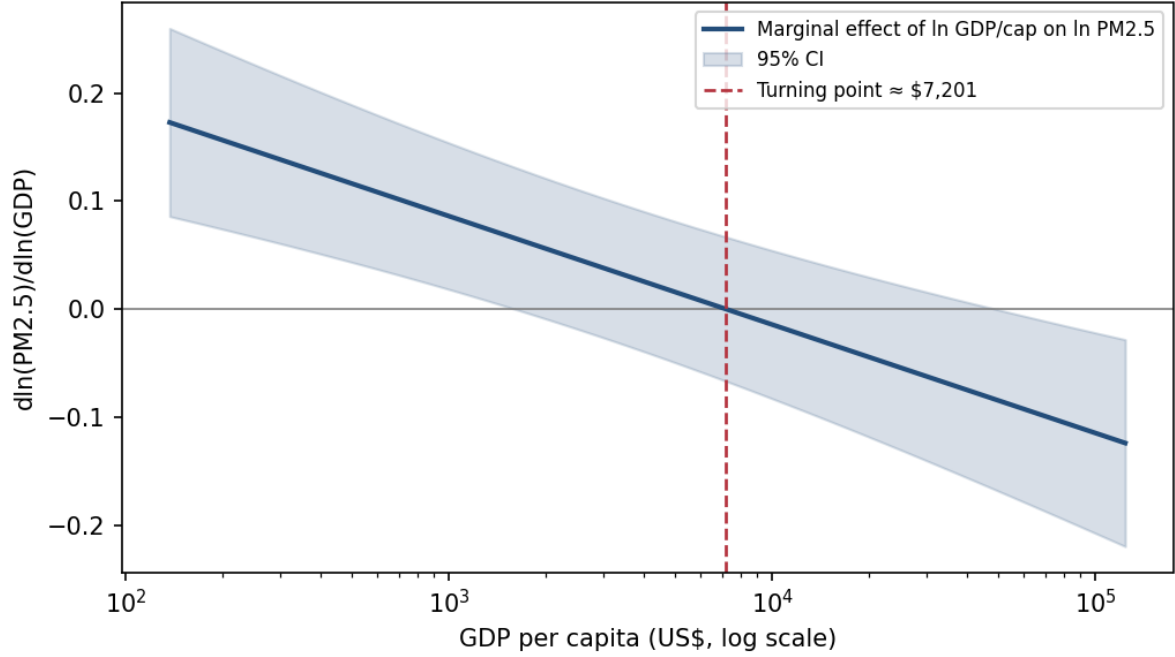


Figure 2: GDP per capita marginal effect on PM2.5 (Model 5)

The marginal effect of income on PM2.5 exposure is positive at low income levels and turns negative after the EKC turning point of \$7,201 GDP per capita. The 95% confidence band around this marginal effect is derived from the parameter standard deviations and their covariance. The band remains entirely above zero at the low-income end and entirely below zero at the high-income end, confirming that the sign reversal is statistically robust. The zero-crossing of the marginal effect (the turning point) remains interior to the data range even under the full width of the confidence band.

To decompose the uncertainty in the turning point, we perturb each coefficient individually by one standard deviation. Shifting  $\beta_1$  alone moves the turning point to [\$3,434 – \$15,104], accounting for most of the width of the formal 95% CI. Shifting  $\beta_2$  alone yields [\$6,637 – \$8,172], confirming that the curvature of the curve is tightly estimated. The uncertainty is concentrated in the location of the turning point along the income axis, not in the curvature of the relationship itself.

Together, these analyses confirm that GDP per capita robustly predicts the direction of change in PM2.5 exposure across development stages, but the exact income threshold at which exposure peaks remains uncertain, driven primarily by the imprecision in  $\beta_1$ .

## VII. Conclusion

In this paper, we investigated whether, and to what extent, GDP per capita predicts annual population-weighted exposure to fine particulate matter across countries, thereby testing the Environmental Kuznets Curve hypothesis against PM2.5 specifically. We introduced theoretical explanations for the relationship between development and pollution that identify a shift to pollution-generating mechanisms due to industrialization and a shift from the agricultural to the industrial sector as driving mechanisms towards initial increases in PM2.5 exposure. Beyond the tipping point, exposure is hypothesized to decrease because of a shift from manufacturing into a service economy, technological development, and demand for clean air. We then test these hypotheses against a dataset consisting of the independent variable GDP per capita and the dependent variable annual population-weighted exposure to PM2.5, covering 174

countries from 2000 until 2019. In the basic OLS model, the only significant coefficient is the linear coefficient, implying that economic structural growth leads to a linear decline in PM2.5 exposure. However, after considering fixed temporal and spatial effects, the initial hypothesized environmental Kuznets curve can be replicated. The turning point is identified at \$7,201 with a wide confidence interval, suggesting it should not be taken at face value by policymakers. From the mediating variables, both trade openness and renewable energy production are significantly negatively related to PM2.5 exposure. Furthermore, these findings provide insight into the mechanisms underlying the observed EKC. The reduction of the EK curvature after the inclusion of control variables in model (6) suggests that structural transformation plays a role in linking pollution exposure to economic development. Given its moral salience, the initial increase in PM2.5 exposure should be tackled through mitigation and adaptation to protect vulnerable groups, both domestically and internationally.

## VIII. References

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